

PRITHVI SINGH KANAUIJA

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EDUCATION

University of Massachusetts Amherst

Sep 2020 - May 2024

Bachelor of Science, Computer Science

Recipient of the Chancellor's Award

Relevant Courses: Computer Systems, Data Structures, Artificial Intelligence, Algorithm Design and Analysis, Machine learning, Probability Theory, Web Programming, Search Engines, Computer Networks, Digital Forensics, Natural Language Processing, Data Management, Operating Systems, Game Programming, Multivariate Calculus, Linear Algebra, Computer Vision, Computer Graphics, Mobile and Wireless Networks

EXPERIENCE

Metadome.ai | Troy, MI

Jun 2023 - Sep 2023

Software Engineer Intern, Full-Stack

TypeScript, React, AWS, Git, Babylon.js, C++, Unity, Unreal Engine

- Developed a virtual showroom with Babylon.js, enabling clients to display over **200 products** to prospective customers.
- Engineered and proposed Proof-Of-Concept XR product demos using **OpenGL**, successfully **securing 2 clients** for the company.
- Achieved a **40% cost deduction** by incorporating serverless architecture to store and load assets using Amazon Web Services.
- Collaborated closely with the front-end and design teams to implement **UI/UX** enhancements using React, ensuring seamless control of the WebGL scenes from the parent website.
- Automated the conversion and integration of 3D-generated assets in Unity to Babylon.js scenes, **reducing the total configuration time by 80%**.

Metadome.ai | Delhi, India

Jun 2022 - Sep 2022

Software Engineer Intern, Computer Graphics

Python, C++, Unreal Engine, Blueprint, ARKit, Maya, Figma, Git, Unity, C#

- Updated a motion-capturing system by using Live Link Face to accurately record and replicate a user's real-time facial movements onto virtual characters, **lowering latency by 25%**.
- **Achieved a 20% reduction** in the rendering time of Metahuman characters across different scenes by conducting scalability tests and identifying optimization opportunities.
- Facilitated the onboarding of **15 new recruits** by providing training on the company's Unreal Engine and VR development pipeline.
- Identified and resolved runtime errors for applications hosted on Oculus Quest headsets, **decreasing launch time by 50%**.
- Designed and implemented HUD and UI features for an Unreal Engine application using C++ and Blueprint.

PROJECTS

Image classification model | Python, TensorFlow, Scikit-learn, Node.js, REST, JavaScript

Jun 2023

- Applied machine learning to create a model that can predict the location of a Google street view image from the game GeoGuessr.
- Employed Convolutional Neural Networks (CNN) to stride over parts of each image and train the model.
- Delivered an accuracy of 67% with 84% of the predictions being within 2,000 miles of the exact location.
- Trained the model using 900,000 images around the world sourced from Google's street view API and Mapillary.

Personalized Spotify statistics web application | TypeScript, JavaScript, Node.js, React, REST, Git

Apr 2023

- Showcased a user's Spotify top tracks, insights, and listening history using React and hosted the web application on GitHub pages.
- Integrated Spotify's OAuth2 authentication and API scopes using Node.js to securely access and manage the user's data.

Educational physics puzzle game | C#, C++, Unreal Engine, Blueprint, Unity, Git

Dec 2022

- Implemented AI algorithms for pathfinding and smooth animations in Non-Playable Characters (NPCs).
- Developed algorithms for the procedural creation of diverse levels, landscapes, and textures using free Unity assets, giving the illusion of a sandbox game environment.
- Designed fluid character animations and particle effects using the Niagara FX plugin.

SKILLS

Languages:

Java, Python, JavaScript, TypeScript, C, C++, C#, PostgreSQL, MySQL, HTML, CSS

Frameworks/Libraries:

Three.js, Babylon.js, TensorFlow, SciPy, Scikit-learn, Keras, Node.js, Express.js, REST, React.js, Pandas, NumPy, Django

Other Technologies/Tools:

Unreal Engine, Unity, Amazon Web Services (AWS), Jupyter, Git, Postman, Xcode, Docker, JIRA, Figma